LOCAL PROJECTIONS TRAINING

Bibliography with DOI and Package–Model Inventory

Timberlake Training | Professor Jamel Saadaoui | 25 August 2026

What this guide contains

(1) A DOI-verified bibliography organized by the econometric themes used in the course; and (2) a source-based inventory of the Stata packages, commands, dependencies, and model families presented in the training materials.

AT A GLANCE

Scope	Inventory
Core course	Time-series LP; panel LP; state-dependent LP; TVP-LP; LP-DiD
Supplementary extensions	Quantile LP; smooth-transition LP; horizon-by-horizon AIPW / doubly robust LP
Primary Stata tools	lpirf; locproj / lpgraph; lpdid; tvpreg / tvpplot
Manual engines	regress; newey; xtreg; qreg; var / irf; xthdidregress, aipw
Software baseline	Stata 18 or later for built-in lpirf; SSC packages plus bundled TVP code

Source basis: course-outline PDF, webinar materials, regression/HDID files, and the amended Timberlake Stata code-and-data archive supplied for this task.

1. Bibliography with DOI

1.1 Foundations, comparison with VARs, and inference

Jordà, Ò. (2005). Estimation and inference of impulse responses by local projections. *American Economic Review, 95(1), 161–182.* https://doi.org/10.1257/0002828053828518

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Plagborg-Møller, M., & Wolf, C. K. (2021). Local projections and VARs estimate the same impulse responses. *Econometrica, 89(2), 955–980.* https://doi.org/10.3982/ECTA17813

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Newey, W. K., & West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica, 55(3), 703–708.* https://doi.org/10.2307/1913610

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Jordà, Ò., Schularick, M., & Taylor, A. M. (2017). Macrofinancial history and the new business cycle facts. *NBER Macroeconomics Annual, 31, 213–263.* https://doi.org/10.1086/690241

1.2 Nonlinear, state-dependent, quantile, and time-varying LPs

Auerbach, A. J., & Gorodnichenko, Y. (2012). Measuring the output responses to fiscal policy. *American Economic Journal: Economic Policy, 4(2), 1–27.* https://doi.org/10.1257/pol.4.2.1

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Inoue, A., Rossi, B., & Wang, Y. (2024). Local projections in unstable environments. *Journal of Econometrics, 244(2), Article 105726.* https://doi.org/10.1016/j.jeconom.2024.105726

Müller, U. K., & Petalas, P.-E. (2010). Efficient estimation of the parameter path in unstable time series models. *Review of Economic Studies, 77(4), 1508–1539.* https://doi.org/10.1111/j.1467-937X.2010.00603.x

Inoue, A., Rossi, B., Wang, Y., & Zhou, L. (2025). Parameter path estimation in unstable environments: The tvpreg command. *Stata Journal, 25(2), 374–406.* https://doi.org/10.1177/1536867X251341170

Koenker, R., & Bassett, G., Jr. (1978). Regression quantiles. *Econometrica, 46(1), 33–50.* https://doi.org/10.2307/1913643

Mignon, V., & Saadaoui, J. (2025). Asymmetries in the oil market: Accounting for the growing role of China through quantile regressions. *Macroeconomic Dynamics, 29, e38.* https://doi.org/10.1017/S1365100524000312

Ginn, W., & Saadaoui, J. (2025). Monetary policy reaction to geopolitical risks in unstable environments. *Macroeconomic Dynamics, 29, e90.* https://doi.org/10.1017/S136510052500015X

1.3 LP-DiD, staggered adoption, and doubly robust estimation

Dube, A., Girardi, D., Jordà, Ò., & Taylor, A. M. (2025). A local projections approach to difference-in-differences. *Journal of Applied Econometrics, 40(7), 741–758.* https://doi.org/10.1002/jae.70000

de Chaisemartin, C., & D'Haultfœuille, X. (2020). Two-way fixed effects estimators with heterogeneous treatment effects. *American Economic Review, 110(9), 2964–2996.* https://doi.org/10.1257/aer.20181169

Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of Econometrics, 225(2), 254–277.* https://doi.org/10.1016/j.jeconom.2021.03.014

Callaway, B., & Sant'Anna, P. H. C. (2021). Difference-in-differences with multiple time periods. *Journal of Econometrics, 225(2), 200–230.* https://doi.org/10.1016/j.jeconom.2020.12.001

Sun, L., & Abraham, S. (2021). Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics, 225(2), 175–199.* https://doi.org/10.1016/j.jeconom.2020.09.006

Sant'Anna, P. H. C., & Zhao, J. (2020). Doubly robust difference-in-differences estimators. *Journal of Econometrics, 219(1), 101–122.* https://doi.org/10.1016/j.jeconom.2020.06.003

Jordà, Ò., & Taylor, A. M. (2016). The time for austerity: Estimating the average treatment effect of fiscal policy. *Economic Journal, 126(590), 219–255.* https://doi.org/10.1111/ecoj.12332

Roth, J., Sant'Anna, P. H. C., Bilinski, A., & Poe, J. (2023). What's trending in difference-in-differences? A synthesis of the recent econometrics literature. *Journal of Econometrics, 235(2), 2218–2244.* https://doi.org/10.1016/j.jeconom.2023.03.008

1.4 Empirical applications used or closely tied to the replications

Mignon, V., & Saadaoui, J. (2024). How do political tensions and geopolitical risks impact oil prices?. *Energy Economics, 129, Article 107219.* https://doi.org/10.1016/j.eneco.2023.107219

Aizenman, J., Ho, S.-H., Huynh, L. D. T., Saadaoui, J., & Uddin, G. S. (2024). Real exchange rate and international reserves in the era of financial integration. *Journal of International Money and Finance, 141, Article 103014.* https://doi.org/10.1016/j.jimonfin.2024.103014

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2. Stata packages, commands, and dependencies

Interpretation: a package is software; a model is an econometric specification. The inventory below deliberately separates the two.

2.1 Core implementation tools

Tool	Status	Primary role	Where used
lpirf	Built in; Stata 18+	Local-projection IRFs for multivariate time series	Day 1 time-series LP; horizon/lag changes; IRF storage and comparison
var, irf	Built in	VAR benchmark, recursive identification, and IRF management	Construct identified shock; compare LP and VAR response paths
locproj	SSC; S459204	Flexible time-series/panel LP front end	Panel LP; interactions; thresholds; state dependence; optional qreg engine
lpgraph	Installed with locproj	Postestimation graphing and multi-path comparison	Combines and labels response paths across models/states/quantiles
lpdid	SSC; S459273	Dynamic DiD through local projections and clean controls	Absorbing and nonabsorbing treatment; reweighting; comparator restrictions
tvpreg, tvpplot	Bundled ado/Mata/help	Time-varying parameter regression and coefficient-path plots	TVP-LP paths by calendar time and horizon; event-conditioned displays
regress, newey	Built in	Manual horizon-by-horizon LP estimation	OLS and Newey–West/HAC inference for overlapping time-series horizons
xtreg, areg	Built in	Panel fixed-effects engines	Manual panel LP and comparison specifications
qreg	Built in	Conditional quantile regression engine	Supplementary quantile LP via locproj, m(qreg)
xthdidregress, aipw	Built in	Heterogeneous DiD / AIPW benchmark	Regression-analysis module; useful benchmark, not itself an LP command

2.2 Installed support packages

Package	Status	Function in the training environment
ftools	SSC	Fast grouped operations required by high-dimensional fixed-effects workflows
reghdfe	SSC	Absorbs multiple fixed effects; compiled in the amended installer
bgshade	SSC	Background/event shading used in time-varying coefficient graphs
moremata	SSC	Additional Mata utilities supporting matrix-intensive routines

2.3 Legacy/reference-only installations

Package	Treatment in the supplied materials
listreg	Installed in the 2025 webinar/reference do-file but omitted from the amended installer
boottest	Commented installation in the regression/LP-DiD source; not required by the amended archive

Package	Treatment in the supplied materials
egenmore	Commented installation in the regression/LP-DiD source; not required by the amended archive

Installation baseline in the amended archive

ssc install locproj; ftools; reghdfe; lpdid; bgshade; moremata. The TVP command files are supplied locally and added to the Stata ado-path by the setup script.

3. Models and estimators presented in the training

3.1 Time-series local projections and benchmarks

Model	Core estimand/design	Implementation in the training
Manual linear LP	Estimate one regression per horizon	regress/newey; coefficients and confidence bands assembled across h
Built-in multivariate LP	Dynamic response of a variable system to a shock	lpirf with chosen lags and step; IRFs stored through irf
Recursive VAR benchmark	Cholesky-ordered structural innovation	var + irf; used to build/validate an identified shock and compare paths
Identified-shock LP	LP response to a shock recovered from the recursive VAR	Generated shock enters lpirf/manual projections as an exogenous driver
Lag/horizon sensitivity	Robustness of response timing, persistence, and precision	Grid of lag lengths and maximum horizons

3.2 Panel local projections

Model	Core estimand/design	Implementation in the training
Fixed-effects panel LP	Average dynamic response using country/unit and time effects	Manual FE regressions and locproj panel syntax; clustered inference
Panel LP with forward-shock controls	Dynamic response net of subsequent shock realizations	Adds future shock terms to probe persistence/feedback
Interaction / marginal-response LP	Response varies continuously with reserves or another moderator	Shock × moderator terms; marginal paths evaluated at relevant values
Threshold / regime panel LP	Responses differ across low/high financial-development or openness regimes	Split-sample and interacted state specifications; threshold robustness
Historical macro panel LP	Panel response using long-run macrofinancial data	Jordà–Schularick–Taylor dataset replication and horizon comparisons

3.3 Nonlinear, distributional, and time-varying extensions

Model	Core estimand/design	Implementation in the training
Constant-parameter LP	Single response path assumed stable over the sample	Baseline against which flexible specifications are compared

Model	Core estimand/design	Implementation in the training
Hard state-dependent LP	Separate IRFs by an indicator-defined regime	Shock × state interactions; full-sample and developed-economy examples
Smooth-transition LP	Regime membership changes continuously through a logistic weight	Supplementary extension; smooth high/low-state response paths, no separate package
Quantile LP	Dynamic response at selected conditional quantiles	Supplementary extension; locproj with m(qreg), typically 20th/50th/80th quantiles
TVP-LP	Impulse-response coefficient path evolves over calendar time	tvpreg/tvpplot; all-horizon, selected-horizon, and event-conditioned paths

3.4 Dynamic treatment effects and LP-DiD

Model	Core estimand/design	Implementation in the training
TWFE event study	Lead/lag coefficients around treatment timing	Used as a diagnostic benchmark; highlights staggered-adoption contamination
Manual LP-DiD	Horizon-specific ATET using newly treated units and clean controls	Long differences; valid comparison sample constructed separately at each horizon
LP-DiD: absorbing treatment	Dynamic effects when treatment persists once adopted	lpdid baseline; reweighted, nocomp, never-treated, and pooled-mean-difference variants
LP-DiD: nonabsorbing treatment	Dynamic effects when treatment can switch on and off	lpdid nonabs() with alternative not-yet-treated comparison restrictions
AIPW / doubly robust LP	Horizon-specific effect combines an outcome model and propensity model	Supplementary manual extension; overlap/common-support and IPW diagnostics; no separate LP package
Built-in HDID AIPW benchmark	Group-time treatment effects robust if either nuisance model is correct	xthdidregress aipw in the regression-analysis module; benchmark rather than LP estimator

4. Quick selection guide

Research need	Recommended training model/tool
Time-series shock; conventional linear response	lpirf or manual LP with newey
Need a VAR comparison or recursive shock	var + irf, then identified-shock LP
Panel shock with fixed effects	locproj or manual FE panel LP
Response differs by a binary regime	State-dependent LP
Response changes smoothly with the state	Smooth-transition LP
Response differs across the outcome distribution	Quantile LP via qreg engine
Response changes over calendar time	TVP-LP via tvpreg/tvpplot
Staggered absorbing policy adoption	LP-DiD via lpdid with clean controls
Treatment switches on and off	lpdid with nonabs()
Want double robustness to nuisance-model misspecification	Manual AIPW/DR LP; compare with xthdidregress aipw when applicable